Hugo Marquerie 27/12/2025

La convergencia débil y en el dual implica la convergencia de la evaluación

Proposición 1. *Sea X un espacio de Banach real, $(x_n)_{n\in\mathbb{N}} \subset X$ y $(L_n)_{n\in\mathbb{N}} \subset X^*$ tales que $x_n \rightharpoonup x$ y $L_n \to L$. Entonces, $L_n(x_n) \xrightarrow{n\to\infty} L(x)$.*

***Demostración*:** Basta ver que $|L_n(x_n) - L(x)| \xrightarrow{n\to\infty} 0$ en $\mathbb{R}$:

$$\begin{aligned}
|L_n(x_n) - L(x)| &= |L_n(x_n) - L(x) + L(x_n) - L(x_n)| = |(L_n - L)(x_n) + L(x_n - x)| \\
&\leqslant |(L_n - L)(x_n)| + |L(x_n - x)| \\
&\leqslant \|L_n - L\| \cdot \|x_n\| + |L(x_n - x)| .
\end{aligned}$$

Como $x_n \rightharpoonup x$, por la `prop-norma-semicontinua-inf-convergencia-debil/`**??**, $\exists C > 0 : \forall n \in \mathbb{N} : \|x_n\| < C$. Como además, $L_n \to L$, $\|L_n - L\| \to 0$. Por tanto, el primer sumando tiende a 0.

Por otro lado, como $x_n \rightharpoonup x$, entonces $L(x_n) \to L(x)$, es decir, $|L(x_n - x)| \to 0$. Por tanto, ambos sumandos tienden a 0, luego $|L_n(x_n) - L(x)| \to 0$. ■

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